Coupling biofilm growth into an existing FEM element

Master's thesis colloquium

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Motivation and objective

- Peri-implantitis affects about one in five implant patients (prevalence 22 % and 19.5 % in two meta-analyses)
- It is driven by multi-species biofilms; a shift towards pathobionts such as *P. gingivalis* goes with disease
- A biofilm is also a mechanical body: where it grows unevenly or against a stiff substrate, it carries stresses

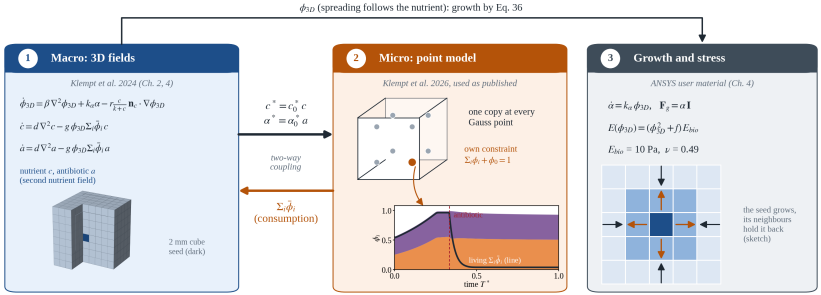
Objective: to compute the species composition of a biofilm, its growth and the resulting stresses consistently within an existing finite element framework.

Point model
species composition at a material point
no spatial resolution, no stress

Growth field
spatial growth and growth-induced stress
single biofilm phase, no species

This work
both, evaluated at every Gauss point of an existing ANSYS element

Approach



Output: ϕ_i, ψ_i at each Gauss point. $\bar{\phi}_i = \phi_i \psi_i$: only living biomass consumes. Assumed, not from a paper: c^*, α^* linear in time, $\Sigma_i \bar{\phi}_i$ one step late, new points take the neighbours' living composition, $\phi_{min} = 0.001$ (converged). x from the data window.

1. Calibration
which of the 15 interactions the data determine (TMCMC, 4 conditions, 3 seeds)

2. Growth law in ANSYS
$\mathbf{F}_g = \alpha \mathbf{I}$ as a user material, verified against an exact solution

3. Coupling
composition reaches the stress only through nutrient and front speed; grid error known

→ The point model runs unchanged at every Gauss point; nutrient and antibiotic come from the fields, and only living biomass consumes them.

Both models from the extended Hamilton principle

Growth field (Klempt et al. 2024)

$$\dot{\phi} = \beta \nabla^2 \phi + k_\alpha \alpha - r \tfrac{c}{k+c} \mathbf{n}_c \cdot \nabla \phi$$
$$\dot{c} = d \nabla^2 c - g\, \phi \sum_i \bar{\phi}_i\, c, \qquad \dot{\alpha} = k_\alpha \phi$$
$$\mathbf{F} = \mathbf{F}_e \mathbf{F}_g, \quad \mathbf{F}_g = \alpha \mathbf{I}, \quad \alpha(0) = 1$$

Stress only from $\mathbf{F}_e$; growth is $\alpha - 1$.

$\phi \in [0,1]$: amount of biofilm; c: nutrient concentration; α: growth variable, growth $= \alpha - 1$; β: diffusion coefficient of ϕ [mm$^2/T^*$]; k_α: growth rate [$1/T^*$]; d, g: nutrient diffusivity [mm$^2/T^*$], consumption rate [$1/T^*$]; T^*: normalised time of Klempt et al. 2024

Point model (Klempt et al. 2026)

$$\dot{\Psi} + \Delta + C \;\rightarrow\; \underset{\dot{\boldsymbol{\phi}}, \dot{\boldsymbol{\psi}}, \gamma}{\text{stat}}$$

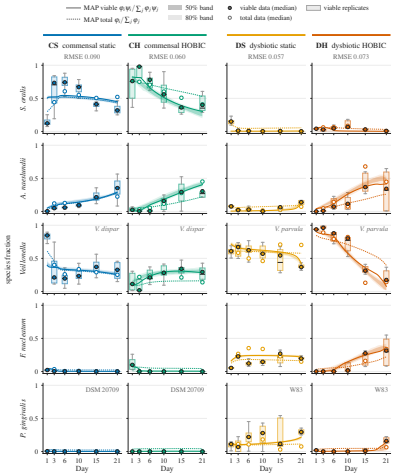
$\Psi = -\frac{1}{2} c^* \bar{\boldsymbol{\phi}}^{\mathrm{T}} \mathbf{A}\, \bar{\boldsymbol{\phi}}$, $\bar{\phi}_i = \phi_i \psi_i$, $C = \gamma(\sum_l \phi_l - 1)$

local form: principle of the minimum of the dissipation potential (Junker et al. 2014, Hackl 1997)

$\Rightarrow$ $\mathbf{A}$ symmetric, $2n+2$ equations per point

ϕ_i, ψ_i: volume fraction and viability of species i; $\bar{\phi}_i$: living fraction; Ψ: free energy, Δ: dissipation function, C: constraint function; c^*: nutrient level; $\mathbf{A}$: interaction matrix; γ: Lagrange multiplier of $\sum_{l=0}^{n} \phi_l = 1$ (ϕ_0: void)

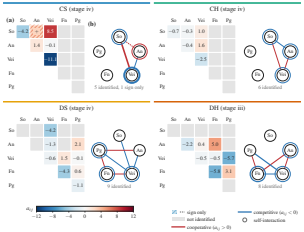
Part 1: Bayesian calibration of the point model

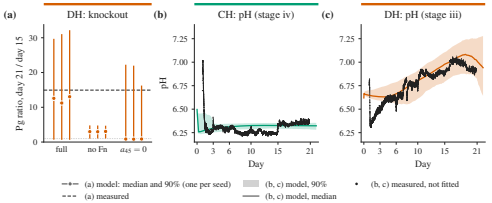


- Heine et al. (2025): five species, four conditions (commensal / dysbiotic, static / flow), days 1–21
- 15 coefficients a_{ij} per condition, effective and undirected
- TMCMC in four stages, three seeds each, with convergence checks
- JAX on the GPU: about 150 times faster than the CPU

Fit: RMSE 0.03 to 0.18; in CS composition and live fraction cannot both be matched

Part 1: what the data determine

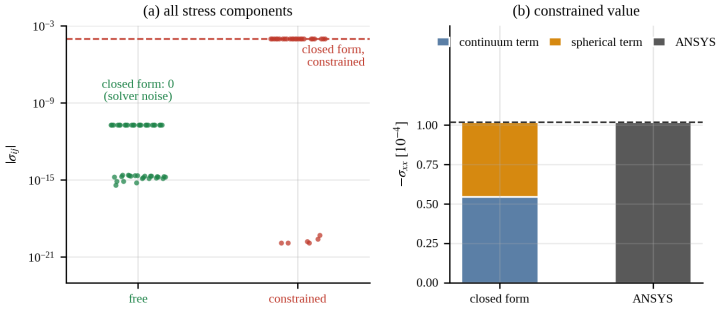




- Identified entries: 4, 6, 8, 9 of 15 (CS, CH, DS, DH)
- Commensal: only the early colonisers; *So–Vei* positive in both
- pH not used in the inference: a check, followed in DH

→ **Testable hypothesis:** without *F. nucleatum*, the late growth of *P. gingivalis* under dysbiotic flow disappears in the model.

Part 2: implementation of the growth law in ANSYS



Material

ANSYS user material, $\mathbf{F}_g = \alpha\mathbf{I}$, neo-Hookean

Stiffness (Klempt et al. 2024)

$E(\phi) = (\phi^2 + f)E$, $E = 10$ Pa, $\nu = 0.49$; $f = 10^{-3}$ (this work)

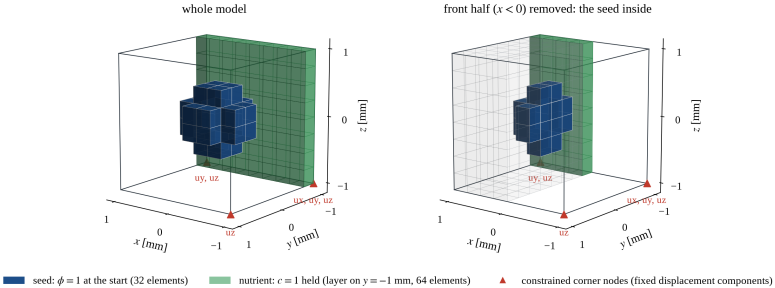
Check (one element)

free: zero stress; constrained: -1.0193×10^{-4} MPa, as predicted

$\rightarrow$ The implementation reproduces the closed-form solutions for free (stress-free) and constrained growth.

Part 2: coupling to the IKM element

ANSYS model: cube of 2 mm, $8 \times 8 \times 8 = 512$ hexahedral elements of 0.25 mm



Model

2 mm cube, 8^3 SOLID185 (B-bar), seed of 32 elements; also 16^3, 24^3

Unchanged

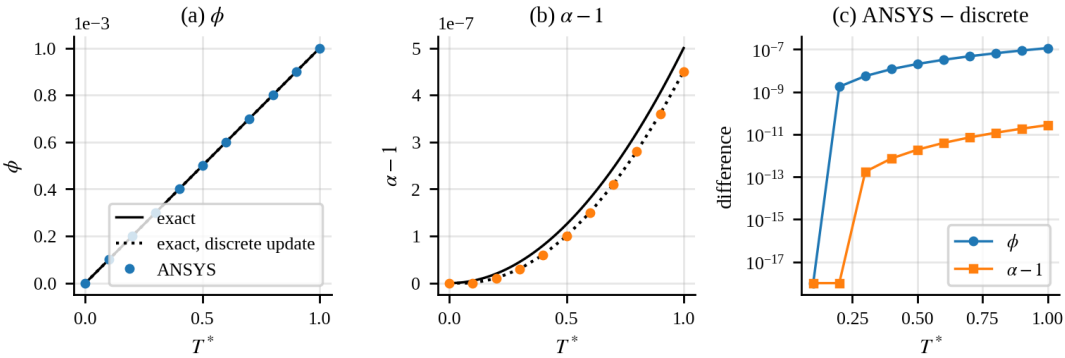
the element's field ϕ (NEM); $\beta = 0.02$ mm$^2/T^*$, converted (assumption)

Added in this work

$\dot{\alpha} = k_\alpha \phi$ at every Gauss point (Eq. 36) $\rightarrow \mathbf{F}_g \rightarrow$ stress

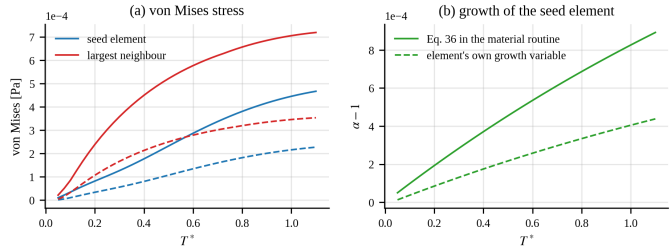
$\rightarrow$ The field formulation of the element is retained; only the growth law at the Gauss points is added.

Part 2: verification against an exact solution



$\rightarrow$ Where ϕ has no gradient, $\phi = \sinh(k_\alpha t)$ and $\alpha - 1 = \cosh(k_\alpha t) - 1$: the coupled ANSYS solution agrees with the exact solution of the time integration to 3×10^{-11} in α.

Part 2: growth-induced stresses with the published stiffness



Left: von Mises stress of the seeded element and its largest neighbour; right: $\alpha - 1$; solid: Eq. 36 at the Gauss points, dashed: the element's own growth variable.

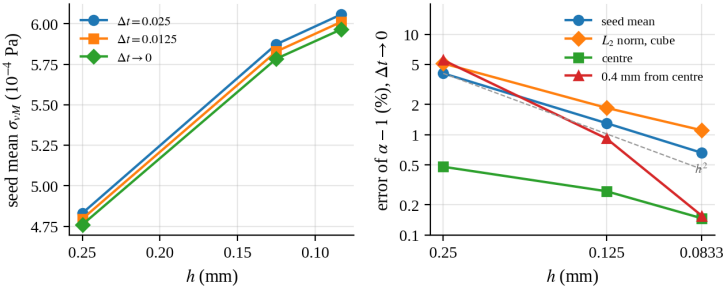
At $T^* = 1.1$ on 16^3 elements:

mean stress, seed	-2.5×10^{-4} Pa
mean stress, first layer	$+6.0 \times 10^{-5}$ Pa
mean stress, second layer	$+1.3 \times 10^{-6}$ Pa
von Mises, seed mean	5.9×10^{-4} Pa
$\alpha - 1$ in the seed	8.9×10^{-4}

Stresses scale with $E = 10$ Pa.

$\rightarrow$ The seed is under compression and the surrounding material under tension, consistent with the ring of tension reported by Klempt et al. 2024.

Part 2: convergence and stability



Time step Δt [T^*]

error about 1 %

Element size h [mm]

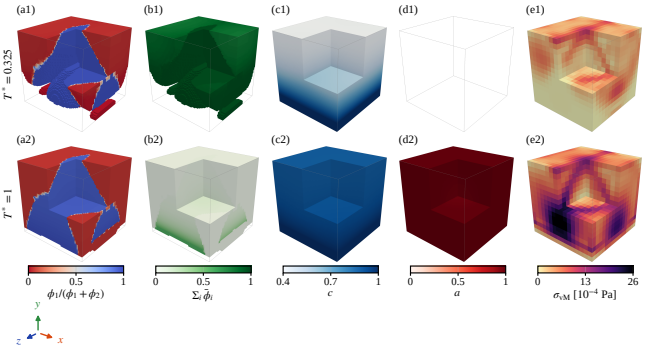
error of $\alpha - 1$ measured against a reference: 0.7 % on 24^3

Stability

limit of the element derived and confirmed; all runs below it (Appendix D)

→ The errors are measured, not estimated, and every run lies within the derived stability limit.

Part 3: two species with nutrient and antibiotic

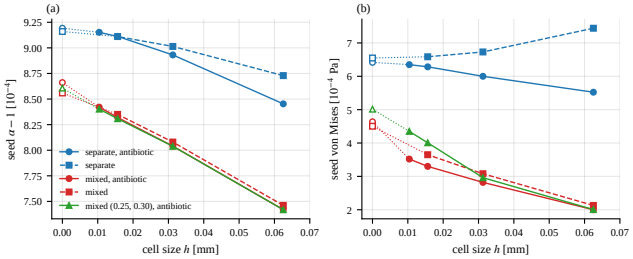


Separate start, $T^* = 0.325$ (top) and 1 (bottom): share of A, $\sum_i \bar{\phi}_i$, c, a, von Mises stress; 64^3, one octant removed

- Synthetic data (Fritsch et al. 2025, case I), all constants from the papers; antibiotic from $T^* = 1/3$, $\alpha_0^* = 100$
- Living fraction of the seed: separate $0.96 \to 0.15$, mixed $0.96 \to 0.04$
- Growth changes by less than 2 % and the converged seed stress by about 2 %: the antibiotic lowers the living fraction, not ϕ_{3D}
- New points take the living composition of their neighbours (assumption)

$\to$ The composition reaches growth and stress only through the nutrient and the front speed $rc/(k+c)$.

Part 3: the front term sets the accuracy of the stress

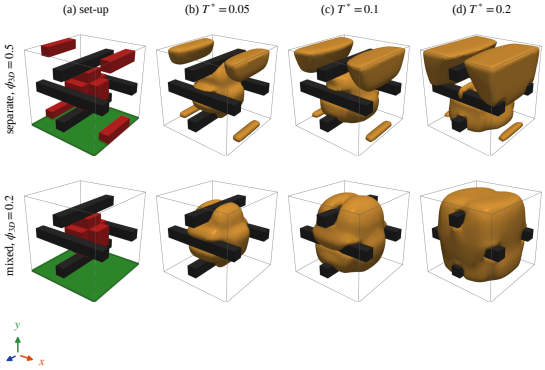


Seed means at $T^* = 1$ against the cell size, all examples, outer step 0.0025; open symbols: extrapolation (three finest grids, Celik et al. 2008; 192^3 with antibiotic)

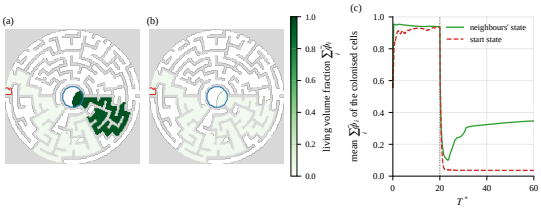
- Front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi$: cell Péclet number ≈ 31 on 16^3; a constant added diffusion gives a quarter of the converged stress, a second-order upwind scheme is needed
- With it, up to 192^3: growth GCI below 4.5 %, separate stress 1.4 %; mixed stress 19 to 41 % with minmod, 4.6 % with the superbee limiter, which keeps the edge sharper (Backup 17)
- The IKM element with upwind gradients: 5.8×10^{-4} Pa (GCI 16.5 %), inside the band

→ The stress needs a second-order treatment of the front term; with it the grid error is known.

Part 3: rigid bars and a maze



Four rigid bars after Klempt et al. 2024, Fig. 9; surface ϕ_{3D} at $T^* = 0.05$ to 0.2



Maze of Klempt et al. 2024, Fig. 11, antibiotic from $T^* = 20$: new points take the neighbours' state (a) or the start state (b); living fraction over time (c)

- Bars restrain the swelling: seed von Mises about ten times higher (32^3, separate start)
- Maze: the rule for new points decides recovery after the antibiotic, 0.35 against 0.04 (assumption)

→ Geometry changes the stress; the start state of new points decides whether the biofilm recovers.

Limitations

- **Verification, not validation**: no measured stress of a growing oral biofilm was available
- **Mechanics one way**: the stress does not act back on growth (as in Klempt et al. 2024), and the composition does not act on the stiffness
- **Front term** $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi$: its speed depends on the nutrient only, so dead biofilm keeps spreading after the antibiotic; the stiffness counts dead biomass
- **Assumptions**: start values of newly reached points, $\phi_{\min}$, the conversion of β; each reported with its sensitivity
- **Calibrated parameters in the FE model**: one earlier one-way run (CH) only; Part 3 uses synthetic data
- **Symmetric A**: no directed effects; only part of $\mathbf{A}$ is determined; different strains in the commensal and dysbiotic data

$\rightarrow$ Each limitation is quantified in the thesis where possible; the results are verified, not yet validated.

Conclusions and outlook

Contributions

1 Calibration: five species, four conditions; which interactions the data determine

2 Growth law: Klempt et al. 2024 in the IKM element, verified to 3×10^{-11}

3 Composition: the point model at every Gauss point, coupled through nutrient and antibiotic; it reaches the stress only through the nutrient and the front speed

Outlook (continuation at Keio University)

- Five calibrated species in the finite element model
- Stress into growth (Eq. 30 of Klempt et al. 2024), composition into stiffness
- A second-order front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi$ in the element; explicit against implicit field update
- Posterior uncertainty carried to the stress

References

- F. Klempt, M. Soleimani, P. Wriggers, P. Junker (2024), Biomech. Model. Mechanobiol. 23, 2091–2113: growth field, Eq. 34/36, Table 2
- F. Klempt, H. Geisler, M. Soleimani, P. Junker (2026), Arch. Appl. Mech. 96, 164: multi-species point model
- P. Junker, D. Balzani (2021), Contin. Mech. Thermodyn. 33, 1931–1956: extended Hamilton principle
- P. Junker, J. Makowski, K. Hackl (2014), Contin. Mech. Thermodyn. 26, 259–268; K. Hackl (1997), J. Mech. Phys. Solids 45, 667–688
- T. Rudolf et al. (2025), Finite Elem. Anal. Des. 249, 104353: Neighbored Element Method
- N. Heine et al. (2025), Front. Oral Health: in-vitro data
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Appendix A: notation

Growth field and mechanics (Klempt et al. 2024)

ϕ, ϕ_{3D}	amount of biofilm (0 = void, 1 = dense)
c	nutrient concentration
α	$\mathbf{F}_g = \alpha\mathbf{I}$, $\alpha(0) = 1$; growth is $\alpha - 1$
$\mathbf{F} = \mathbf{F}_e\mathbf{F}_g$	elastic / growth split
k_α	growth rate ($10^{-3}\, T^{*-1}$)
β	diffusion of ϕ (0.02 mm$^2/T^*$, converted)
d, g	nutrient diffusivity, consumption
r, k	front growth parameter, half-velocity constant
E, ν	10 Pa, 0.49 (Klempt 2024, Table 2)
T^*	normalised time

Point model (Klempt et al. 2026)

ϕ_i, ϕ_0	volume fraction of species i, void
ψ_i, $\bar{\phi}_i$	viability, living fraction $\phi_i\psi_i$
A_{ij}	symmetric interaction matrix
c^*, α^*	nutrient, antibiotic level
η_i	dissipation parameters
γ	multiplier of $\sum_l \phi_l = 1$

Coupling (this work)

$\phi_1/(\phi_1 + \phi_2)$	composition (share of species 1)
s	point model advances $s\,\Delta t$ per substep
ϕ_{cap}, ϕ_{min}	cap on the amount, start threshold
$\lambda = \beta\Delta t/h^2$	stability measure

Appendix B: growth field and mechanics

Kinematics. $\mathbf{F} = \mathbf{F}_e \mathbf{F}_g$ with isotropic $\mathbf{F}_g = \alpha \mathbf{I}$. Stress comes only from $\mathbf{F}_e$. A body that can grow freely stays stress free; stress appears where growth is constrained.

Material. $E(\phi) = (\phi^2 + f)\, E$, $E = 10$ Pa, $\nu = 0.49$ (Klempt 2024, Table 2; their material is incompressible), $f = 10^{-3}$; decks in mm and MPa, stresses quoted in Pa.

Field equations (Klempt et al. 2024, Eqs. 34–36, without the bars):

$$\dot{\phi} = \underbrace{\beta \nabla^2 \phi}_{\text{diffusion}} + \underbrace{k_\alpha\, \alpha}_{\text{local source}} \underbrace{- \|\nabla \phi\| \frac{r\, c}{k + c} \mathbf{n}_{\nabla\phi} \cdot \mathbf{n}_{\nabla c}}_{\text{front, inactive in ANSYS}}, \qquad \dot{c} = d \nabla^2 c - g\, \phi\, c, \qquad \dot{\alpha} = k_\alpha\, \phi$$

- ϕ and c are fields of the element, updated explicitly with the Neighbored Element Method; α is a state variable at the Gauss points
- Consumption $g\phi c$ as in the authors' implementation (printed: $g\phi$)

Appendix C: point model and one coupled substep

Point model (Klempt et al. 2026), per species i, no space:

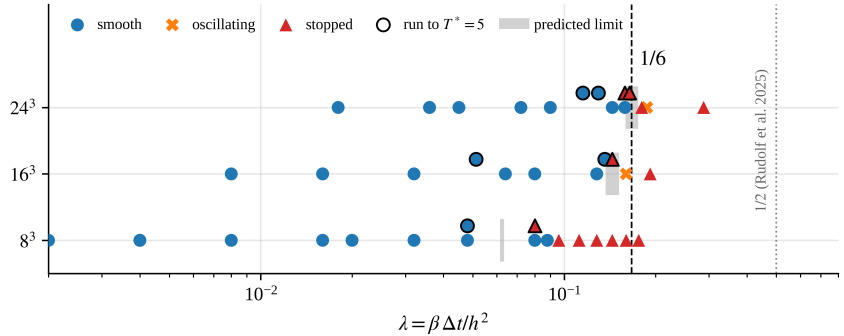
$$(\eta_{\phi,i} + \eta_i \psi_i^2)\,\dot{\phi}_i + \eta_i \phi_i \psi_i\,\dot{\psi}_i = c^* \psi_i \textstyle\sum_j A_{ij} \phi_j \psi_j - \gamma - \text{(barrier)}$$
$$\eta_i\,(\phi_i \psi_i\,\dot{\phi}_i + \phi_i^2\,\dot{\psi}_i) = c^* \phi_i \textstyle\sum_j A_{ij} \phi_j \psi_j - \alpha^* b_i\,\psi_i - \text{(barrier)}$$

with $\sum_i \phi_i + \phi_0 = 1$; the barrier keeps ϕ_i, ψ_i in $(0,1)$; implicit Euler and Newton, $\Delta t = 10^{-4}$.

One substep of the coupled ANSYS run

1. The element updates its field ϕ
2. At each Gauss point: $\alpha_{n+1} = \alpha_n + k_\alpha \phi\,\Delta t$, $\mathbf{F}_g = \alpha \mathbf{I}$, stress and tangent
3. Where $\phi_{3D} \geq \phi_{\min}$: advance the point model by $s\,\Delta t$ ($s = 0.1$) with $c^* = c_0^* c$ and $\alpha^* = \alpha_0^* a$, linear in time; store ϕ_i, ψ_i and $\sum_i \bar{\phi}_i$
4. The element advances ϕ_{3D} (Eq. 34 with the front term $-r \frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi$), c and a, consumed by $g\,\phi_{3D}(\sum_i \bar{\phi}_i)$

Appendix D: stability of the explicit field update



$\lambda = \beta\Delta t/h^2$; grey: the limit from a Bloch analysis of the NEM stencil on the lattice of the Gauss points (thesis, Appendix D). $h^2/(2\Delta t) \geq \beta$ of Rudolf et al. 2025 is the one-dimensional limit $(1/2)$; the seven-point stencil gives $1/6$; the NEM fit with its weight width $\beta^* = 0.2$ mm fixed in millimetres gives 0.06 on 8^3 and, depending on which equidistant points enter the 30 neighbours, 0.12–0.14 on 16^3 and 0.12–0.16 on 24^3. Runs to $T^* = 5$ (black edge) confirm it; runs to $T^* = 1.1$ are too short to show an instability near the limit.

Backup 1: stationarity conditions of the point model

$\dot{\Psi} + \Delta + C \to$ stat with respect to $\dot{\phi}_i$, $\dot{\psi}_i$, $\dot{\phi}_0$ and γ, where

$$\Psi = -\tfrac{1}{2} c^* \bar{\boldsymbol{\phi}}^{\mathrm{T}} \mathbf{A}\, \bar{\boldsymbol{\phi}} + \tfrac{1}{2} \alpha^* \boldsymbol{\psi}^{\mathrm{T}} \mathbf{B}\, \boldsymbol{\psi}, \qquad \Delta = \tfrac{1}{2} \dot{\bar{\boldsymbol{\phi}}}^{\mathrm{T}} \boldsymbol{\eta}\, \dot{\bar{\boldsymbol{\phi}}} + \tfrac{1}{2} \dot{\boldsymbol{\phi}}^{\mathrm{T}} \boldsymbol{\eta}\, \dot{\boldsymbol{\phi}}, \qquad C = \gamma\Big(\sum_{l=0}^{n} \phi_l - 1\Big)$$

gives, with $\bar{\phi}_i = \phi_i \psi_i$ and $\dot{\bar{\phi}}_i = \dot{\phi}_i \psi_i + \phi_i \dot{\psi}_i$,

$$\begin{aligned}
0 &= -c^* \psi_i (\mathbf{A}\bar{\boldsymbol{\phi}})_i + \eta_i \big(\dot{\phi}_i \psi_i^2 + \bar{\phi}_i \dot{\psi}_i + \dot{\phi}_i\big) + \gamma, \\
0 &= -c^* \phi_i (\mathbf{A}\bar{\boldsymbol{\phi}})_i + \alpha^* b_i \psi_i + \eta_i \big(\dot{\psi}_i \phi_i^2 + \bar{\phi}_i \dot{\phi}_i\big), \\
0 &= \gamma + \dot{\phi}_0, \qquad 0 = \textstyle\sum_{l=0}^{n} \phi_l - 1.
\end{aligned}$$

- $2n+2$ equations; implicit Euler and Newton, $\Delta t = 10^{-4}$; a logarithmic barrier keeps ϕ_i, ψ_i, ϕ_0 in $(0,1)$
- The second term of Δ is needed: the first alone gives a singular system
- Only the symmetric part of $\mathbf{A}$ enters $\Psi \Rightarrow \mathbf{A} = \mathbf{A}^{\mathrm{T}}$

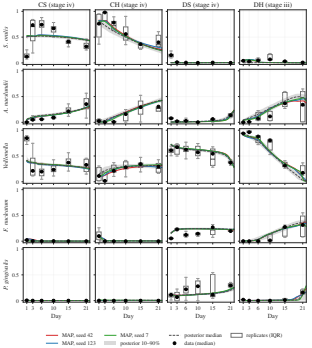
Backup 2: symmetric interactions compared with an asymmetric model

	gLV (L-BFGS-B)	Hamilton TMCMC
interaction matrix	asymmetric, 25	symmetric, 15
parameters	30	20
RMSE (four conditions)	0.012–0.032	0.027–0.179
uncertainty	point estimate	posterior, three seeds
$\sum_i \phi_i = 1$	enforced afterwards	exact

- The asymmetric model fits better, but has more parameters (30) than observations (25); its out-of-sample error was not assessed
- Directed effects (one species feeding another) cannot be represented; stated as a limitation

$\rightarrow$ Symmetry costs accuracy of the fit, but gives a posterior and exact conservation.

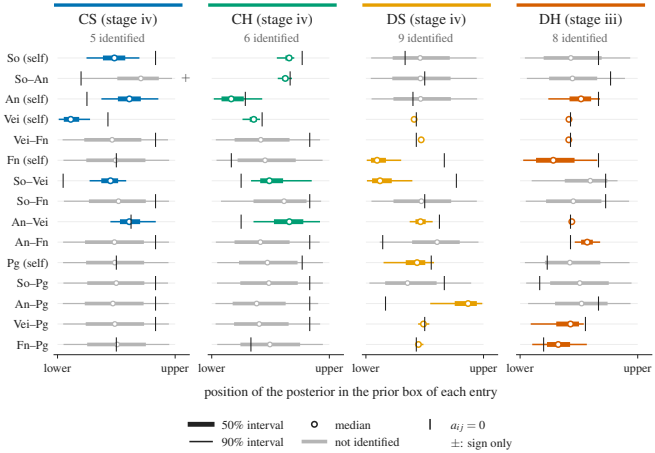
Backup 3: four stages and three seeds



Stages: (i) composition; (ii) narrowed box; (iii) with the live fraction, full box; (iv) narrowed box. Accepted only if recomputed likelihoods agree, tempering reaches $\beta = 1$, every particle moves enough, and the seeds agree in $\max \ln \mathscr{L}$ and in the medians.

$\rightarrow$ Several runs that looked converged failed these checks and were repeated; DS and DH are from stage (iii).

Backup 4: posterior distributions of the interactions



- Marginal posteriors of the 15 entries, drawn by their position in the prior box
- Identified: posterior SD below 0.8 of a uniform distribution on the box
- Sign only: 90 % interval excludes zero

Backup 5: sensitivity of the coupled examples

- **Time step** (outer step $0.025 \to 0.0025$): composition and $\alpha - 1$ within 3 and 0.3 %; stress $-6\,\%$ (separate) and $+15\,\%$ (mixed) at 0.025, so the stress is taken at the small step
- **Start values of new points**: neighbours' living composition against the seed composition, a few percent in the cube; in the maze of Klempt et al. 2024 recovery after the antibiotic (0.35) or not (0.04)
- **Antibiotic over time** (constant, pulses, sine, same dose): growth 0.1 %, stress 0.8 %; the living fraction follows the dose (pulsed: recovers $0.17 \to 0.30$)
- **Nutrient diffusivity**: with $\dot{c}$ and d of Table 2 the fields adjust in $4 \times 10^{-8}\,T^*$; only a value eight orders smaller changes the result
- **Stress mesh**: 16^3 against 64^3 elements, 0.2 %

$\to$ The composition is robust; the stress depends on the time step and on how the front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi$ is discretised.

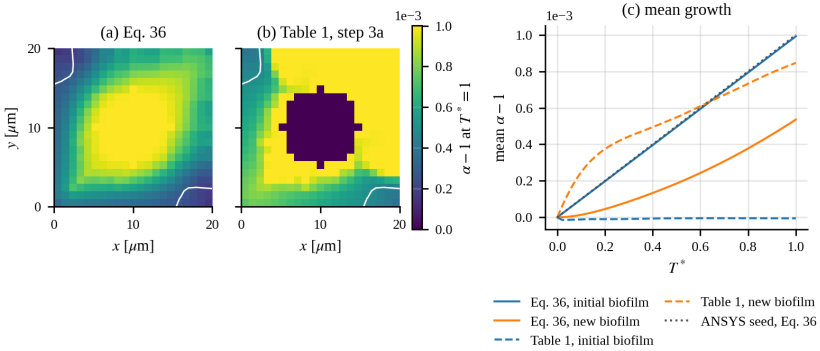
Backup 6: sensitivity to the diffusion coefficient β

Mean stress in the seed at $T^* = 1.1$ [Pa]

β [mm$^2/T^*$]	0.005	0.01	0.02	0.05	0.1
8^3	-2.1×10^{-4}	-2.7×10^{-4}	-2.5×10^{-4}	-1.3×10^{-4}	-5.6×10^{-5}
16^3	-3.9×10^{-4}	-3.4×10^{-4}	-2.5×10^{-4}	$-1.2 \times 10^{-4\,\dagger}$	$-3.8 \times 10^{-5\,\dagger}$

- $\beta = 0.02$: Table 2 of Klempt et al. 2024 ($\beta = 2$) converted to the 2 mm cube (assumption)
- ϕ diffusing out of the seed lowers its growth near the surface; the step in α between inside and surface loads the seed
- For $\beta \geq 0.02$ the diffusion length $\sqrt{\beta T^*} \geq 0.14$ mm is resolved by the mesh
- † before the correction of the surface points

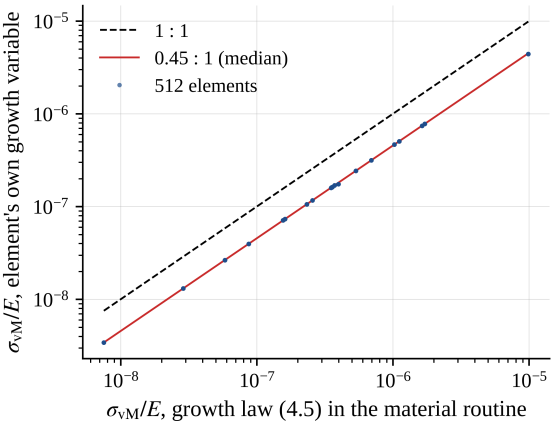
Backup 7: the two growth laws of Klempt et al. 2024



The paper gives $\dot{\alpha} = k_\alpha \phi$ (Eq. 36) in the text and a different update in Table 1, step 3a. On the reproduced field of test case 4.1 they differ: a seed full from the start grows only under Eq. 36.

→ I follow Eq. 36, as the text of the paper and the element do.

Backup 8: Eq. 36 against the element's own growth variable



- Whole model, 512 elements: the two von Mises fields have the same pattern, cosine similarity 1.0000
- The element's variable gives 0.455 times the stress of Eq. 36, which is $\frac{1}{2} \times \frac{10}{11}$: the mean over both species fields, one of them unused, and a lag of one substep
- With Eq. 36 at the Gauss points, the growth is counted once per increment

$\rightarrow$ Same spatial pattern; the difference in magnitude is fully explained by how the element's variable is formed, not by the mechanics.

Backup 9: mesh and incompressibility

- Python, same mechanical problem on 8^3, 16^3, 32^3 linear hexahedra: seed averages on 8^3 are 1.6–1.75 times those on 16^3 and 32^3, which agree with each other; corner peaks do not converge (singular corners) $\Rightarrow$ seed averages are quoted, not peaks
- Mean-dilatation (B-bar) integration, as SOLID185 with KEYOPT(2)= 0: mean stress -3.8 and -4.1×10^{-5} Pa for $\nu = 0.49$ and 0.499 on 8^3 $\Rightarrow$ no volumetric locking
- Full $2 \times 2 \times 2$ integration would overestimate the mean stress by a factor of 3.7 on 8^3 and 2.3 on 16^3 at $\nu = 0.49$
- ANSYS: 16^3 and 24^3 with every boundary condition mapped by position; 32^3 exceeds the memory of the workstation

$\rightarrow$ The quoted seed averages are mesh-converged on 16^3 and 24^3; B-bar avoids locking at $\nu = 0.49$.

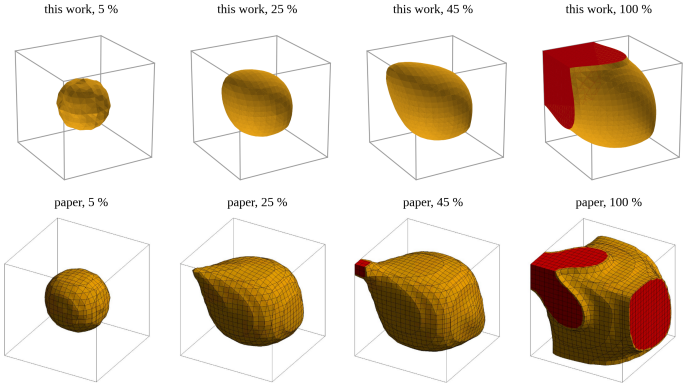
Backup 10: correction of the surface points of the element

- In the element, ϕ at the surface points was not updated and stayed 0: every face acted as $\phi = 0$, where the model has zero flux
- Corrected: each surface point takes the distance-weighted mean of its neighbouring Gauss points; all thesis runs repeated

runs	seed von Mises	seed pressure	$\alpha - 1$
$\beta = 0.02$ (8^3, 16^3, one and two species)	$\leq 0.01\,\%$	$\leq 0.13\,\%$	unchanged
$\beta = 0.05$, 8^3	$+0.3\,\%$	$-3.5\,\%$	$0.02\,\%$
$\beta = 0.1$, 8^3	$+6.7\,\%$	$-30\,\%$	$0.5\,\%$

$\rightarrow$ The main results change by less than $0.5\,\%$; only runs in which ϕ reaches the faces change noticeably, and their corrected values are used.

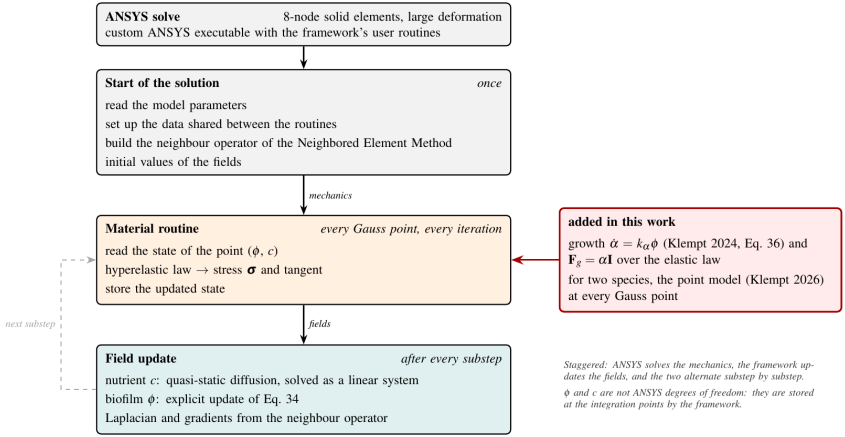
Backup 11: reproduction of Klempt et al. 2024, test case 4.1



Python reproduction with the parameters of Table 2; consumption $g\phi c$ as in the authors' implementation (printed: $g\phi$), front term $-r\frac{c}{k+c}\,\mathbf{n}_c \cdot \nabla\phi$ with an isotropic share.

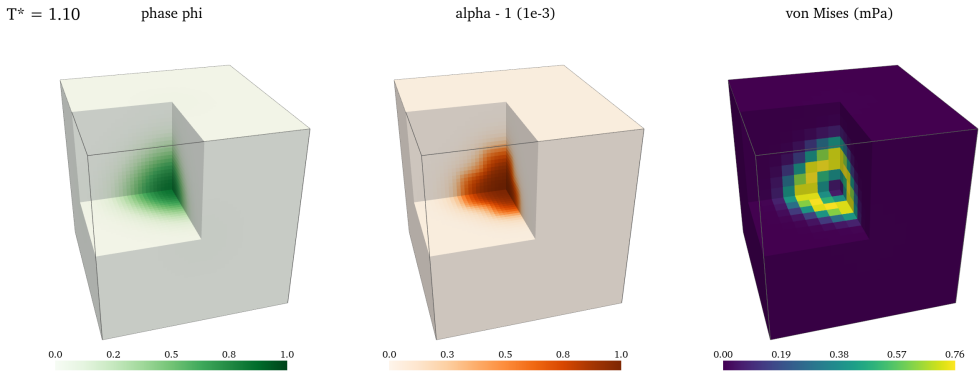
→ Case 4.1 is reproduced with stated departures from the printed equations (thesis, Appendix C).

Backup 12: solution loop of the element framework



→ The material routine is called at every Gauss point and iteration; the fields ϕ and c are advanced after each substep.

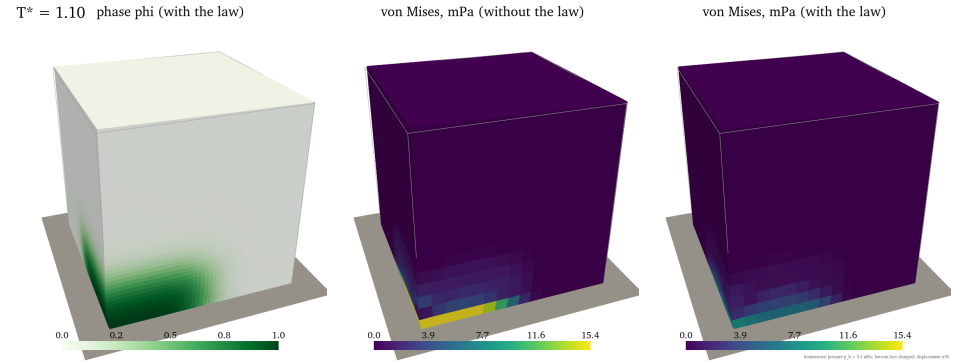
Backup 13: the verification problem in 3D



Seed in the centre of the 2 mm cube, one octant cut away; $\beta = 0.02\,\mathrm{mm}^2/T^*$, $k_\alpha = 10^{-3}/T^*$, $E = 10\,\mathrm{Pa}$. Field: reference solution on 48^3; stress: 16^3. Video: `assets/anim_wp3_seed_3d.mp4`.

$\rightarrow$ Seed von Mises $6.19 \times 10^{-4}\,\mathrm{Pa}$ at $T^* = 1.1$, within $0.2\,\%$ of the value in the thesis; the stress forms a ring at the edge of the seed.

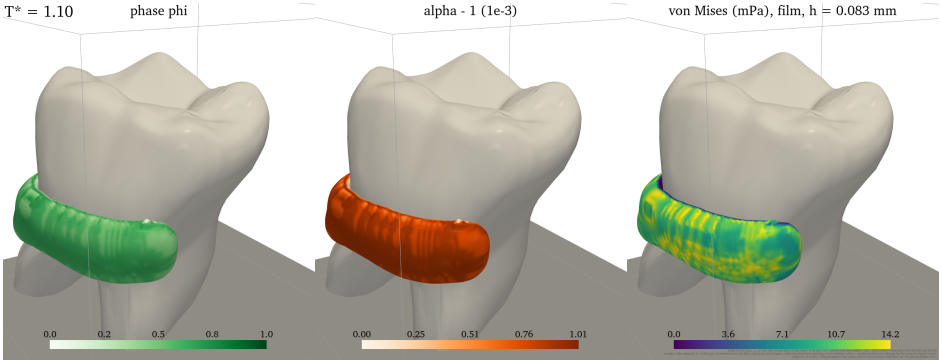
Backup 14: outlook, stress at a wall and a homeostatic-pressure law



Not part of the thesis: the same model with the seed in a corner on a clamped face, without and with growth scaled by $g = \min(1, \max(0, 1 - p/p_h))$; $p_h = 3.3$ MPa chosen by me. Video: `assets/anim_law_3d.mp4`.

→ In the biofilm next to a fixed surface the stress is 10 to 12 times the seed stress of the centre case (band mean, converged to 1 %); limiting growth by pressure removes the concentration. Continuation at Keio.

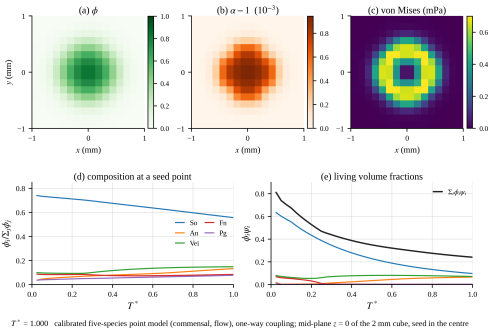
Backup 15: outlook, a biofilm on a real molar



Not part of the thesis: the same growth model on a lower first molar at real size (public surface model, CC BY 4.0), a 0.5 mm film near the gum line. Resolved with six elements across the film, the film mean von Mises stress is 3.3 mPa ($\pm 1\,\%$), highest in the 0.1 mm next to the tooth.

$\rightarrow$ Thin films on real teeth need a film-resolving mesh; with it the stresses converge. Continuation at Keio.

Backup 16: the calibrated five-species model in the element, over time

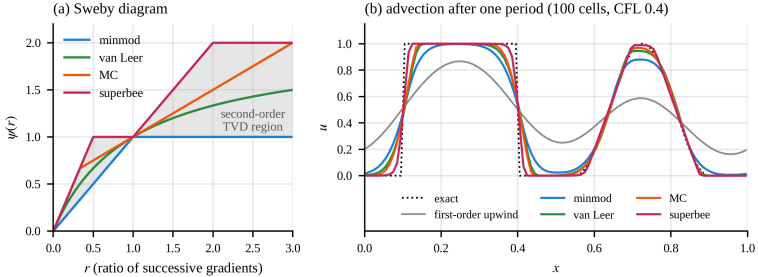


$T^* = 1.000$ calibrated five-species point model (commensal, flow), one-way coupling; mid-plane $z = 0$ of the 2 mm cube, seed in the centre

The run of Section 4 (commensal flow condition, final MAP, Day-1 start, one-way coupling) on the finer 16^3 mesh, 40 output times up to $T^* = 1$. At $T^* = 1$: seed mean $\alpha - 1 = 6.56 \times 10^{-4}$, von Mises 5.74×10^{-4} Pa (8^3 in the thesis: 6.75×10^{-4}, 4.68×10^{-4} Pa); shares within 0.01 of the 8^3 run. Video: `assets/anim_cal5_ch.mp4`.

→ *S. oralis* falls from 0.74 to 0.56 and the living fraction from 0.81 to 0.24, while growth and stress do not depend on the composition (one-way coupling).

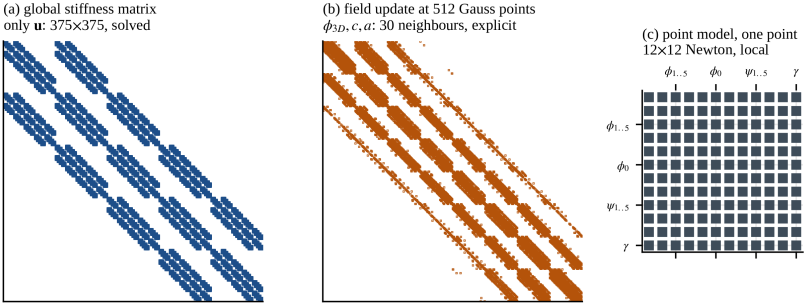
Backup 17: slope limiters for the front term



(a) Limiter $\psi(r)$ against the ratio r of successive gradients; grey: the second-order TVD region of Sweby (1984), bounded by minmod below and superbee above. (b) Textbook test, not the biofilm model: $u_t + u_x = 0$, 100 cells, MUSCL reconstruction, upwind flux, SSP Runge–Kutta 3, CFL 0.4, one period. Minmod smears the edges and clips the smooth peak to 0.88; superbee keeps the edges sharpest and slightly steepens the smooth peak.

$\rightarrow$ The front term $-r\frac{c}{k+c}\mathbf{n}_c \cdot \nabla\phi$ moves a sharp edge; superbee keeps it sharp, so the mixed stress converges on coarser grids than with minmod.

Backup 18: what is solved where



Patterns drawn for 4^3 elements. (a) ANSYS solves only the displacements (3 per node) with the stiffness matrix. (b) The element updates ϕ_{3D}, c and a at its Gauss points explicitly, each from its 30 neighbours (NEM): a sparse product, no solve. (c) The point model has no spatial derivative, so its 12 unknowns are internal variables of each Gauss point, solved there by Newton's method (backward Euler, exact Jacobian, LU), like a plastic strain.

$\rightarrow$ Species add no unknowns to the global matrix: they are solved point by point, inside the material routine.